

Question about environmental impact of AI and / or 3D Art

Posted by u/Grouchy_Ad_4750 • 0 points • 6 comments

I have question for anti ai people who rally against ai because of its impact on environment. if you compare it with 3D art they are really similar:

- \- Both require large amounts of (ideally gpu power)
- \- Both can be selfhosted and run on consumer hardware
- \- AI has datacenters (with gpus) and 3D art can be done on render farms (which is datacenter with gpus)
- \- It could be argued that rendering is even more resource intensive than inference but lets not split hairs and say that they are same

My question is this. Knowing this what restrictions would you like to levy on AI and would you be willing to extend this on 3D art as well? If not why?

Comments

u/mrlegoboy • 1 points • 2025-12-10 10:57 UTC

it doesn't use that much power, not anymore. you used to be able to get away with saying that the models took tons of energy to train but now that they are trained that's just wrong. you can run these models on an average gaming pc.

u/mrlegoboy • 1 points • 2025-12-10 10:59 UTC

and the models are just going to keep getting better and faster as more research is done on how to train and run them. that's just the reality of the situation.

u/JawnStaymoose • 1 points • 2025-12-08 00:06 UTC

Okay. Comparison to 3D art:

- Rendering a 3D scene uses hundreds of watts for a few minutes.
- Training a frontier AI model uses millions of watts for weeks.

It's not even the same universe of energy use.

AI is using a stupid amount of money and power right now.

****How much money?****

- Microsoft, Google, Amazon, Meta are each dropping \$30–70B every year just on AI datacenters.
- Microsoft (OpenAI): ~\$50B/yr on AI infra, plus the rumored \$100B “Stargate” cluster by 2028.
- Google: ~\$50–70B/yr to expand TPU datacenters and rebuild older ones for AI.
- Amazon (AWS): ~\$90B over 2024–26 for Trainium/Inferentia clusters + new regions.
- Meta: \$30–35B/yr, mostly GPU clusters for Llama + ranking systems.
- Total industry spend over the next few years: half a trillion dollars.

This is basically “build a new internet” money. Basically the largest industrial buildout in modern history, rivaling telecom and energy.

****How much power?****

- A full AI megacluster can pull 400–1,000 MW (basically a small nuclear plant).
- Global AI load today = 10–15 gigawatts.
- By 2030 = 30–50 gigawatts (what an entire country uses).

**** The Why it's a problem?**

Beyond the massive amounts of energy and resources requests, for infra, chip creation, etc, we're approaching a choke point.

- Model growth = diminishing returns → more energy per unit improvement.
- Energy availability becomes the limiting factor, not chips.
- Utilities can't keep scaling at the same pace.
- Some companies are openly exploring nuclear-powered datacenters because the growth curve is unsustainable otherwise.

u/Sudden_List_2693 • 1 points • 2025-12-08 16:32 UTC

People have to learn to think, even if AI did the comparison if you mess up your question, that'll be pointless

u/Grouchy_Ad_4750 • 1 points • 2025-12-08 05:57 UTC

Yes training AI takes huge amount of compute effort but so does rendering animated movie for example. If one frames renders few minutes you can imagine how long it takes to render animated scene where you need at least 24 frames per second.

Big studios use render farms and smaller ones rent compute time on them. For example here [<https://x.com/DisneyAnimation/status/1135944096776740865>] (<https://x.com/DisneyAnimation/status/1135944096776740865>) is video of rendering single frame of zootopia on disneys in house render farm.

I guess I have few questions:

\- If it is issue of scale do you object to local usage of AI? Since you can run AI on your own computer and don't need datacenter for that it shouldn't be an issue

\- If you object to all AI do you think 3D artists should stop using AI powered denoisers which are currently used for faster renders and save lots of compute time? (but they are AI and needed to be trained)

Apart from that I fully agree that efficiency should be key in large scale projects like AI. This could either be achieved by better technologies (dedicated low powered chips for inference) or requirements for energy sources (renewable, nuclear, etc...)

u/NegativeKitchen4098 • 3 points • 2025-12-07 23:05 UTC

In all likelihood, the environmental benefit of gen AI is a net positive because of efficiency gains. It's far better for the environment to use gen AI to edit some text and finish it in 10 minutes than working on writing a passage for an hour. It's far better for the environment to do some special effect with AI rather than blow something up with a practical, real life, effect. It's far better for the environment to save some existing footage with gen AI than to have crew go back out and reshoot it.

u/Substantial_Order141 • 3 points • 2025-12-07 19:11 UTC

The comparison is reasonable on hardware, but the environmental impact isn't just about what hardware is used — it's about scale and frequency of use. Most individual 3D artists aren't running massive render jobs constantly, while large AI models are often trained once at huge scale and then used millions of times per day in centralized data centers. Training a large AI model can consume the equivalent of hundreds to thousands of MWh of electricity, while most 3D rendering workloads are project-based and time-limited. After training, AI inference is cheaper per use, but the total environmental footprint scales very fast because of how often it's used. So the concern isn't "GPUs are bad," but industrial-scale, always-on compute versus more occasional, task-based compute. Reasonable restrictions most critics talk about aren't banning AI, but things like renewable energy requirements, transparency in energy usage, and efficiency standards. Those ideas could logically apply to large render farms too, not individual artists. The fair distinction is scale and centralization, not the existence of GPU use itself.

u/Grouchy_Ad_4750 • 3 points • 2025-12-07 19:28 UTC

Thank you for writing thoughtful answer :)

So if one would say self host the AI on his or hers computer you wouldn't have a problem with it? (at least not with environmental impact)

I fully agree that better technologies (renewables, better hw for inference, etc...) are needed and since I've dabbled with 3D art and 3D programming I've seen how much improvement can few

years of technological progress make on that field.

Also I agree think training is most resource intensive part but I believe it is comparable to rendering large animated movies (think pixar) which then also can harm environment by distribution (netflix isn't lightweight either) + most 3D renderers nowadays use AI for denoising output of path tracer for previews etc... Which is one of the exciting developments in the space since before that you've just had to throw more compute at render...